Building Data Apps In Python With Streamlit

Session 4: Deployment

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Course Agenda
(Big Picture)

1. Session 1: Setup & Basics
2. Session 2: "The Loop": UI & Graphics
3. Session 3: Organization: UI & Code
4. Session 4: Deployment

Now you've created an interesting app, and worked hard to make it easy to work with both for users and developers. Let's deploy it to the internet so that others can use it!

Exercise: Deployment?

Work with a partner.

Ask them

- *Have you ever deployed an app to the internet?*
  - *If yes, was it complicated / hard?*
  - *If no, what do you think the process will be like?*

4 minute exercise

Exercise: Deployment?

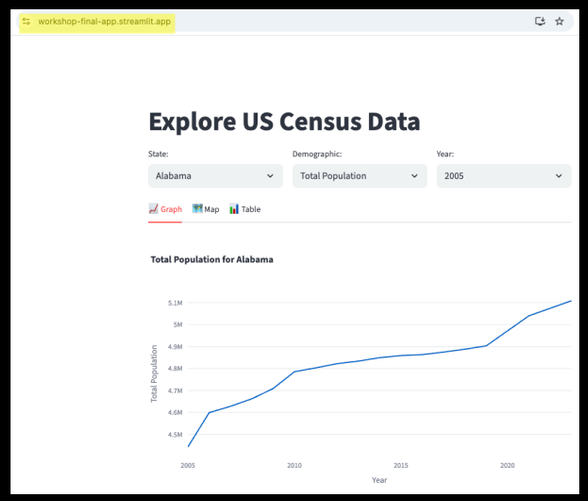
Work with a partner.

Ask them

- *Have you ever deployed an app to the internet?*
  - *If yes, was it complicated / hard?*
    - When I worked at Electronic Arts, we "shipped" the code on physical discs.
    - I worked at HubPages and Trulia (websites), deployment was handled by someone else. It was a "black box" for me.
    - In R, I deployed Shiny apps. The process there was a bit more complex than it is for Streamlit.
  - *If no, what do you think the process will be like?*

Deployment Overview

- Free via Streamlit Community Cloud: https://streamlit.io/cloud
- Share URL with anyone
- Example: https://workshop-final-app.streamlit.app/
- Tied to a github repo



Deployment

- Get the code into Github
- Tell Streamlit Community Cloud about the Repo

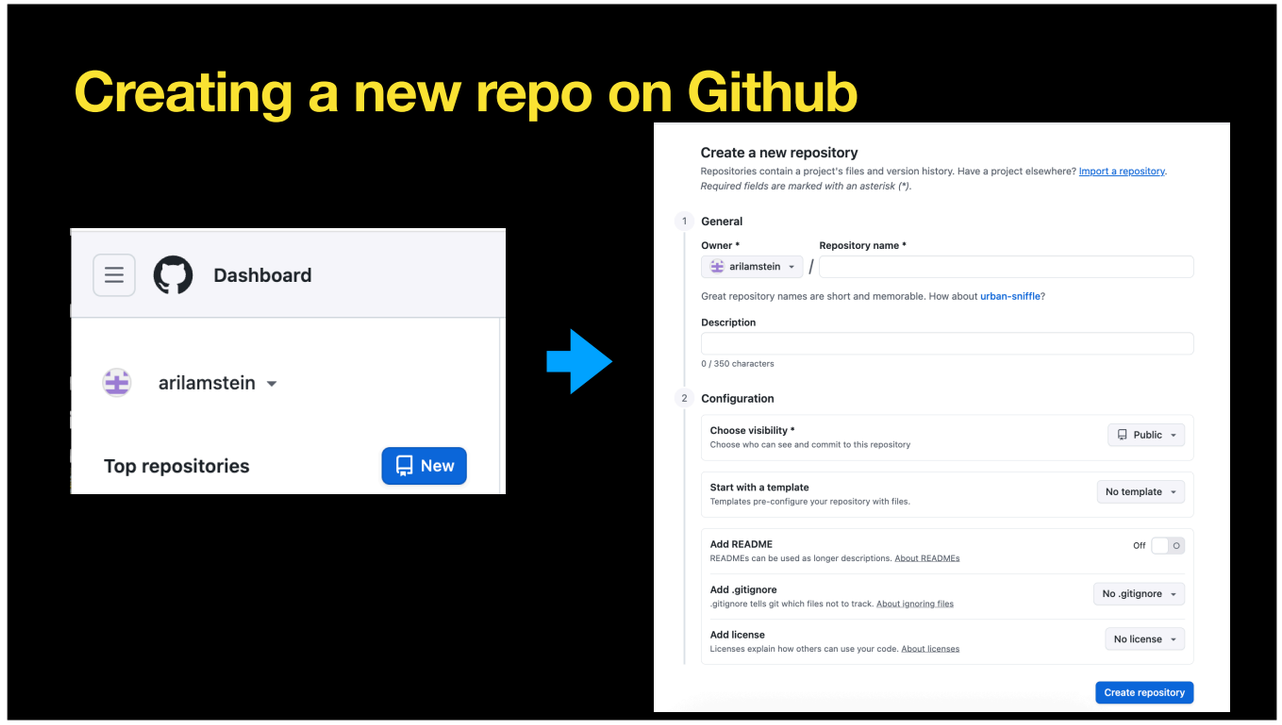
Deploying a Streamlit app is a 2-step process. The first step is to get the code into Github. Then we need to tell Streamlit about the repo.

If you have a github account, please login to it now. If you *don't* have a github account, please let me know and we can create it together.

Deployment

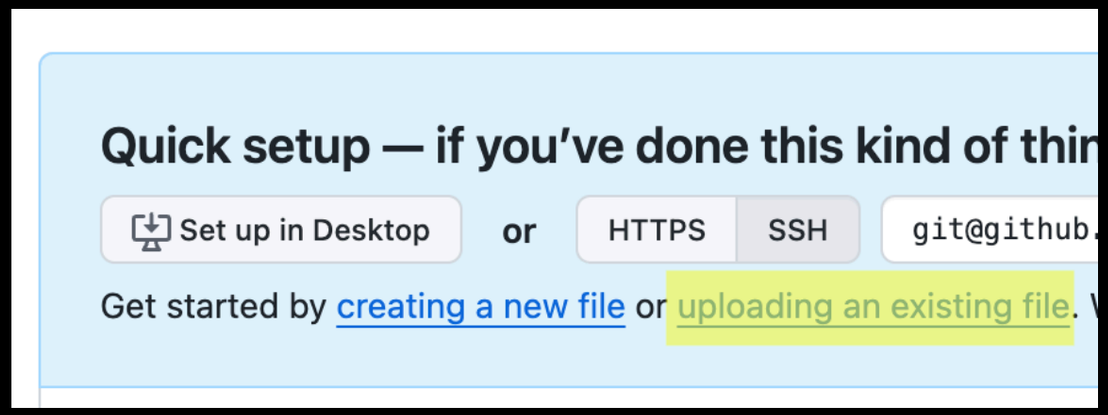
- Get the code into Github
  - Create a new respository
  - Upload files to it
- Tell Streamlit Community Cloud about the Repo

"Getting the code into Github" is itself a 2-step process.
The first step is to create a new repository. Let's look at that.



Your home screen on github might look different than mine. But in the upper left I see a button that says "New". When I click it I get this screen. For the name, enter something like "streamlit-workshop". Leave everything else as it is, and then click "Create repository".

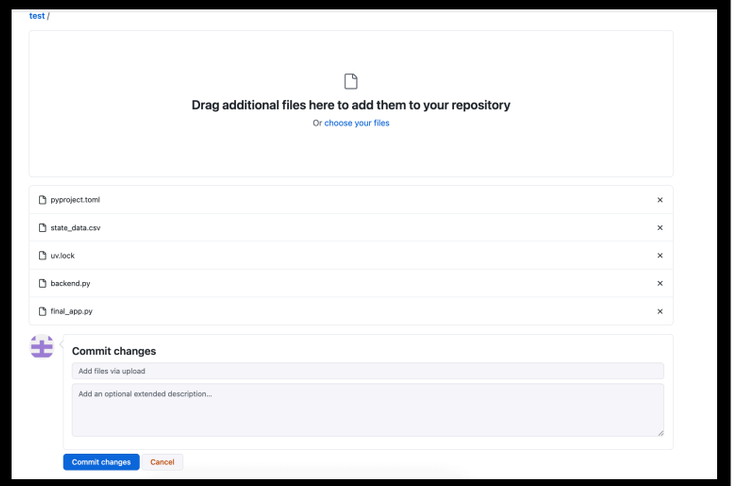
"Upload an existing file"



You should see an option that says "upload an existing file". Click it.

Files to Upload

1. ./python-version
2. /pyproject.toml
3. /uv.lock
4. 3-organization/state_data.csv
5. 3-organization/backend.py
6. 3-organization/layout_app.py



Your new repo needs to have 6 files.

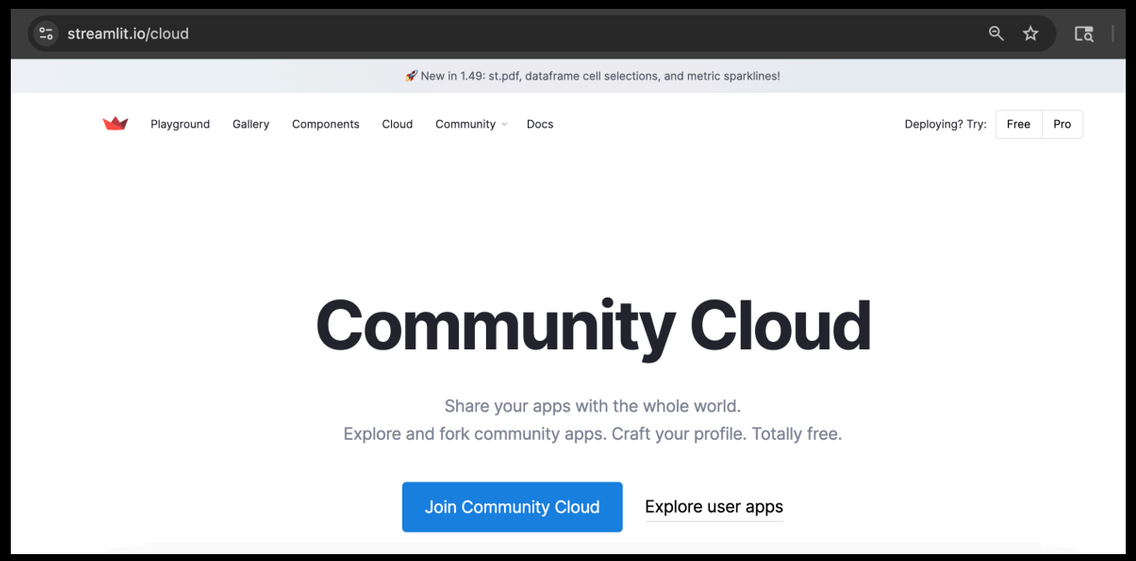
The first 3 are the uv files we spoke about in the first section. These are in the main directory in your repo. Streamlit Cloud uses these files to recreate your Python environment (version and packages).

The other files your repo needs are for the app itself: the data it needs, the ui module and the backend module.

Deployment

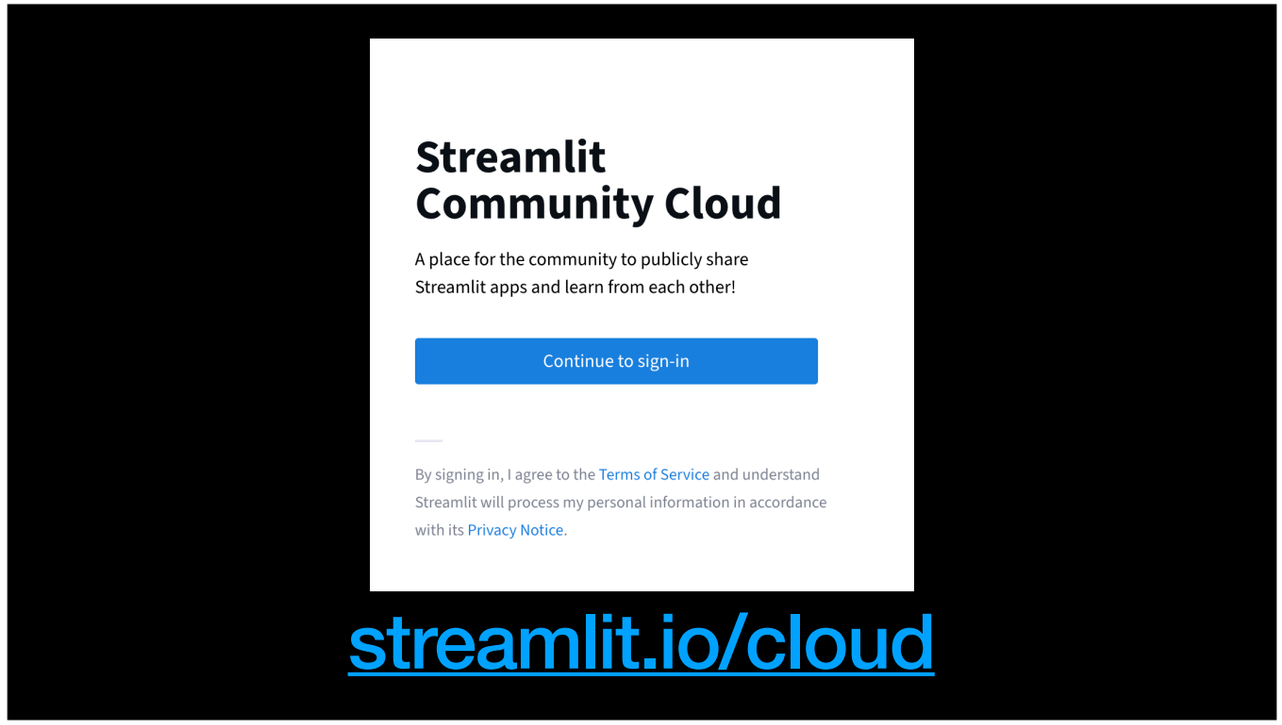
- Get the code into Github
- Tell Streamlit Community Cloud about the Repo

In the last step we made the source code for your app publicly accessible. Now let's tell Streamlit Cloud how to access it.

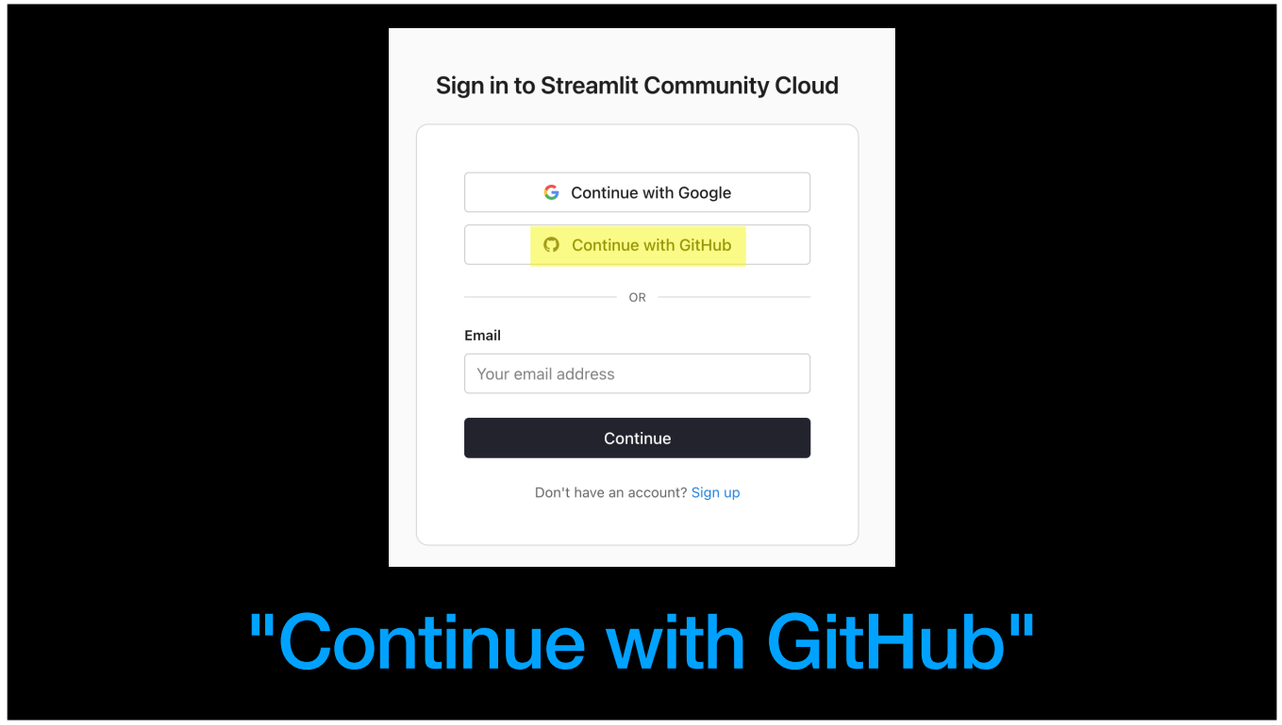


streamlit.io/cloud

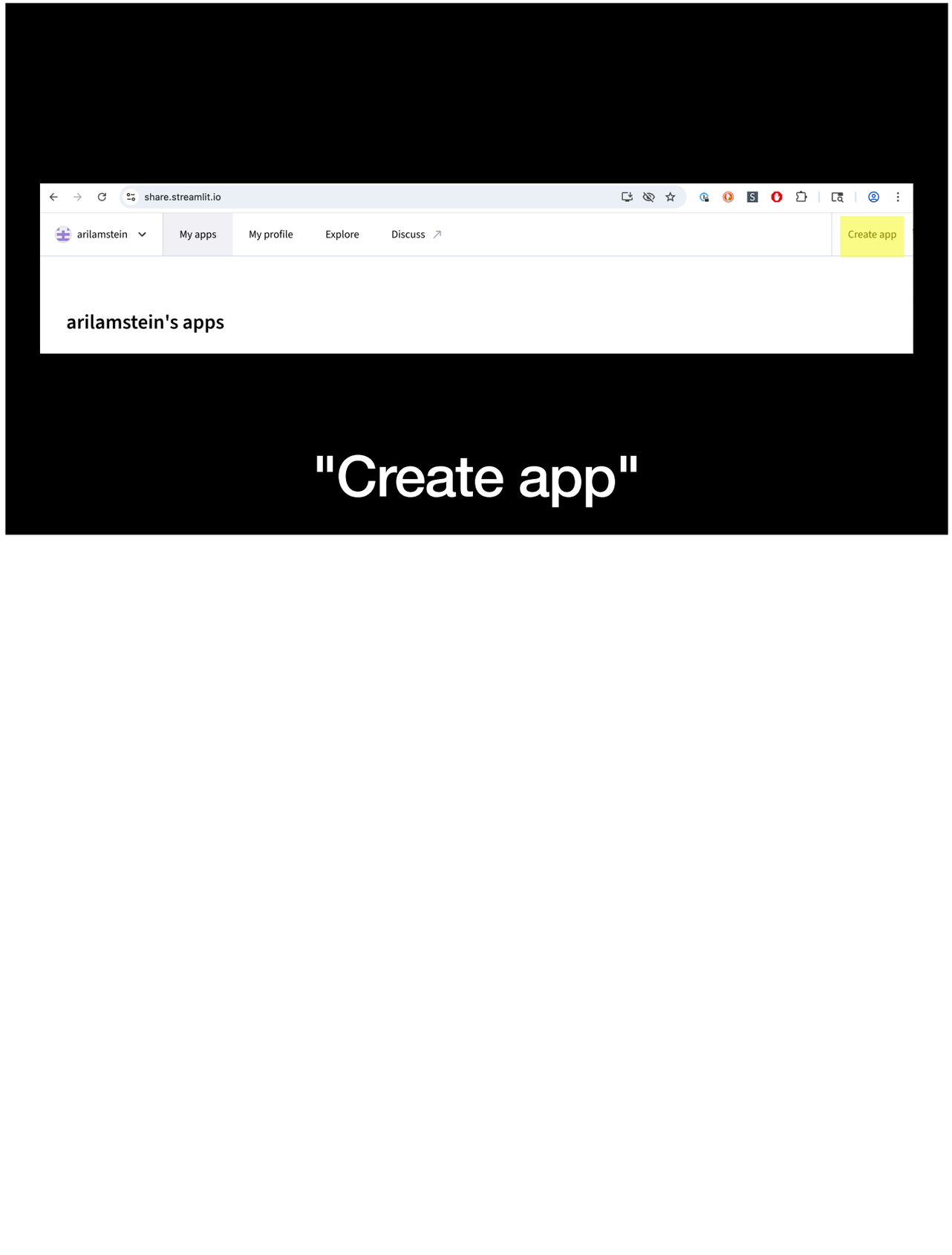
Please visit streamlit.io/cloud and click "Join Community Cloud"



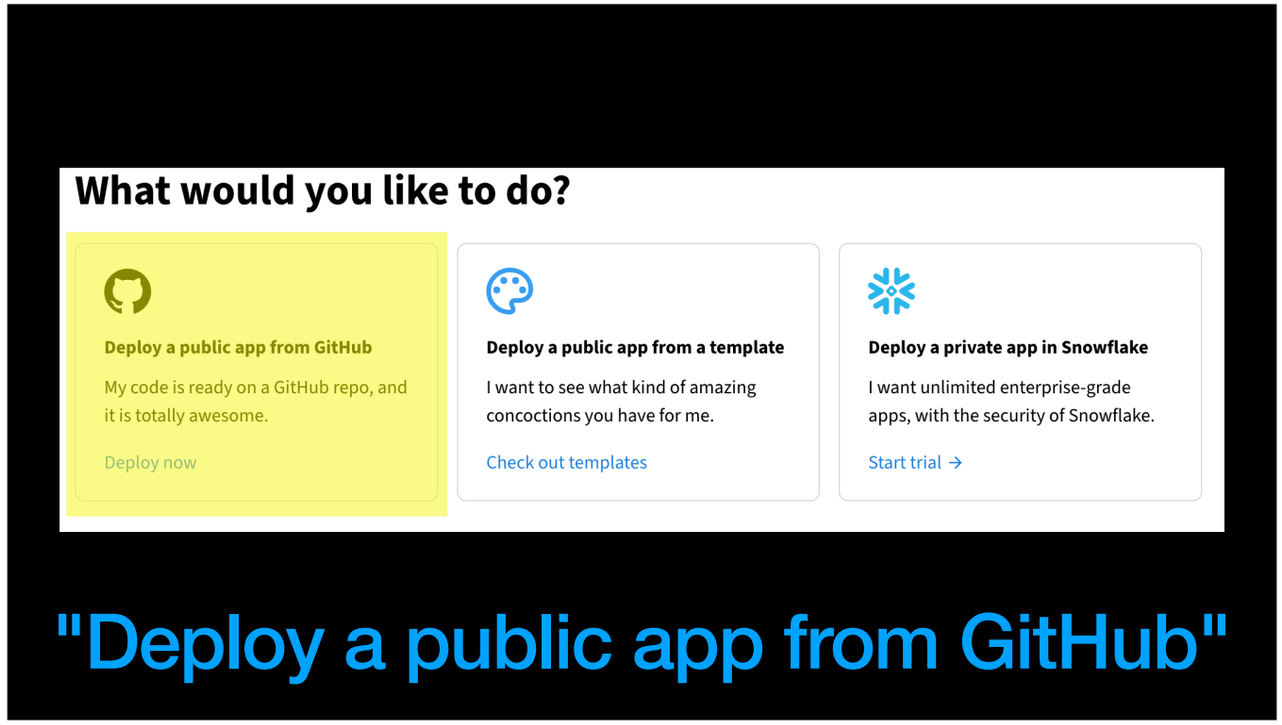
Click "Continue to sign-in"



Click "Continue with GitHub"



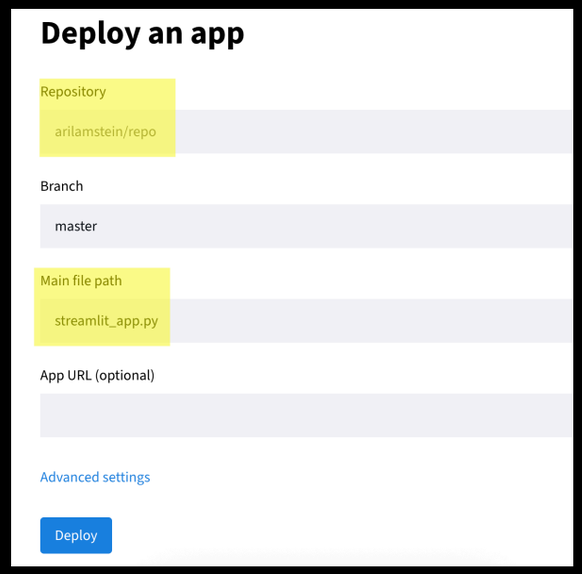
Click "Create app"



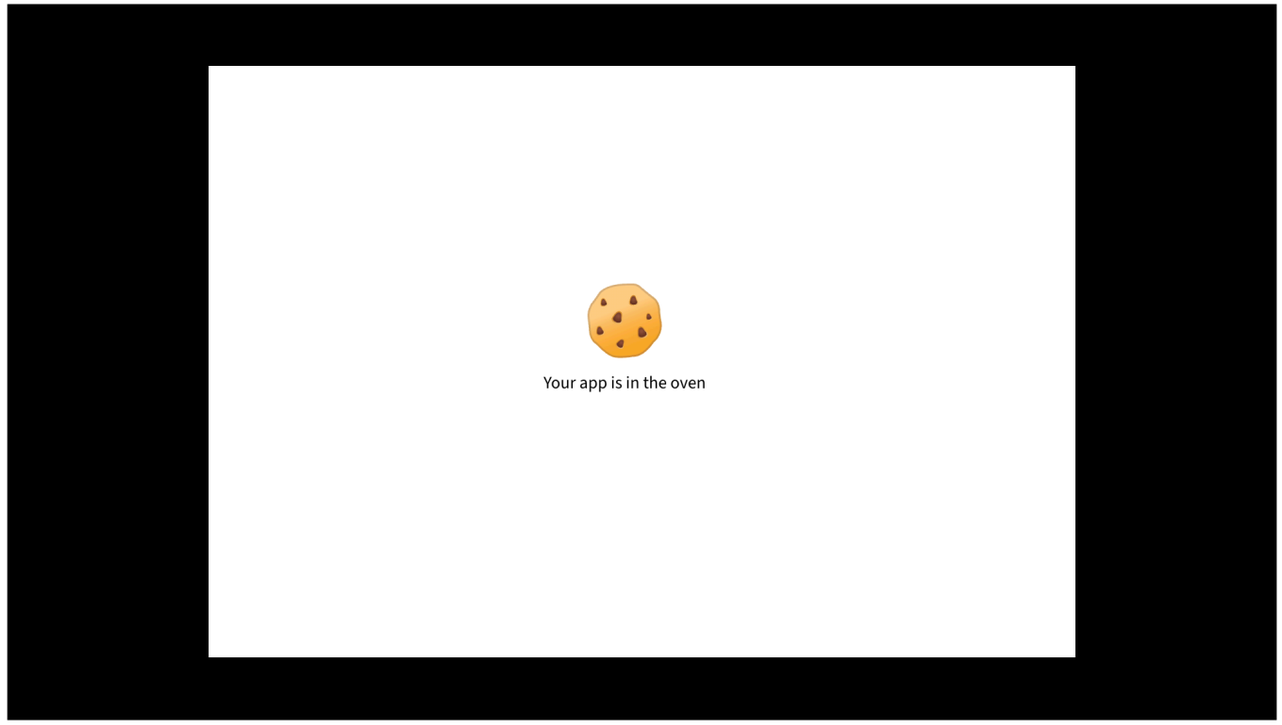
Click "Deploy a public app from GitHub"

Select your Repo

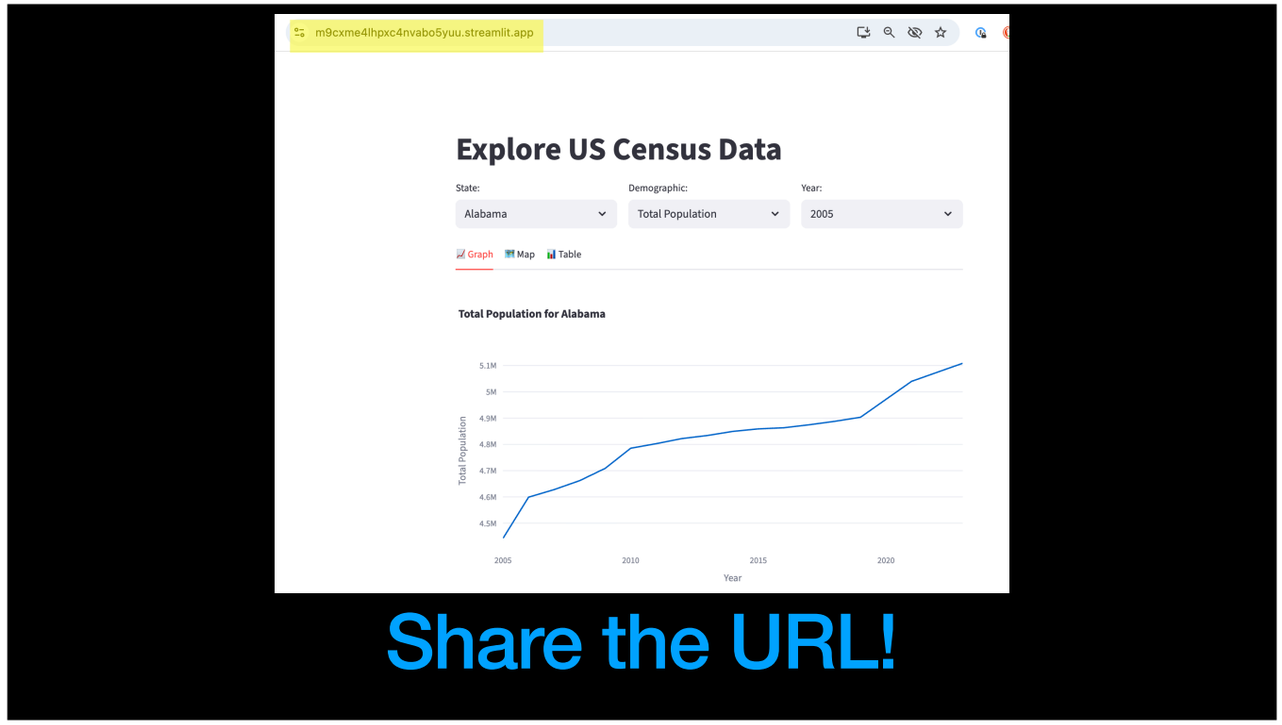
1. Repository box: Copy the URL for your new repo
2. Main file path box: Select your app file (ex. final-app/final_app.py)
3. Click Deploy



I recommend using **final-app/final_app.py** as the app you deploy.



Your app is in the oven

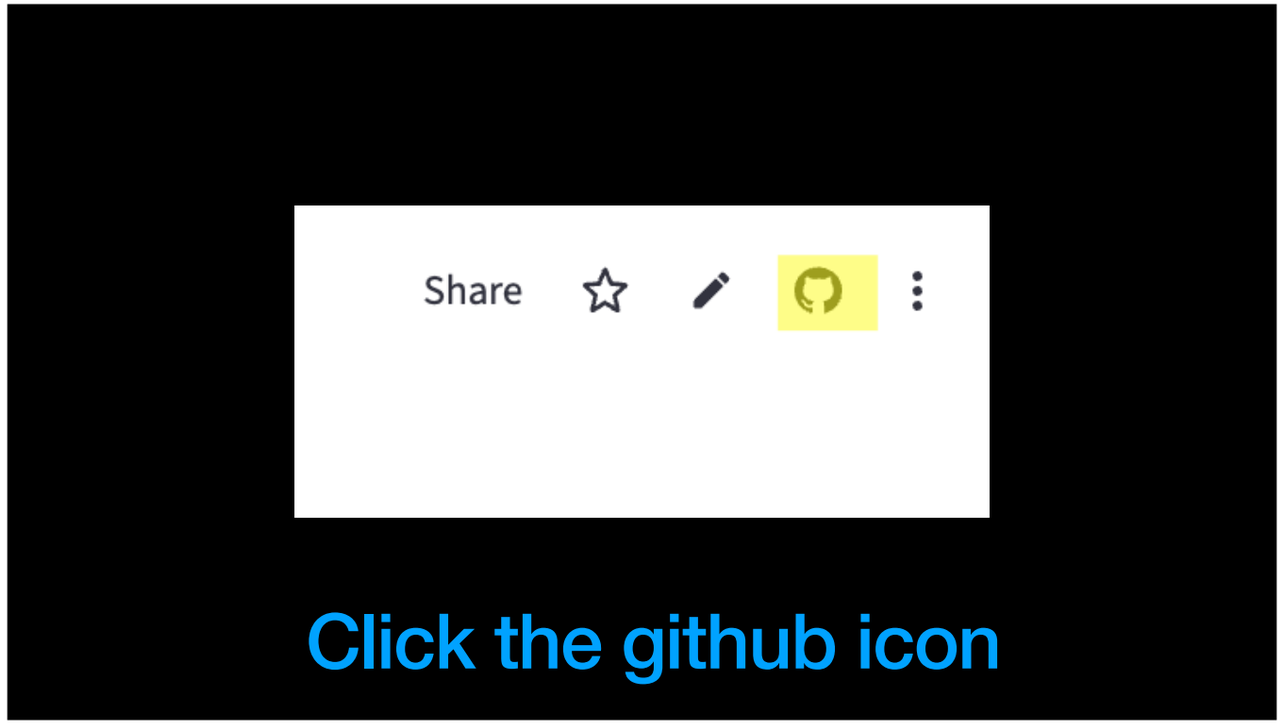


-Test the app to make sure it works just like it did on your local machine.
-Then share the URL with friends, family & colleagues. In my personal experience, sharing the apps I've built has changed how I approach software development.
-Save the URL for your app - you'll need it later

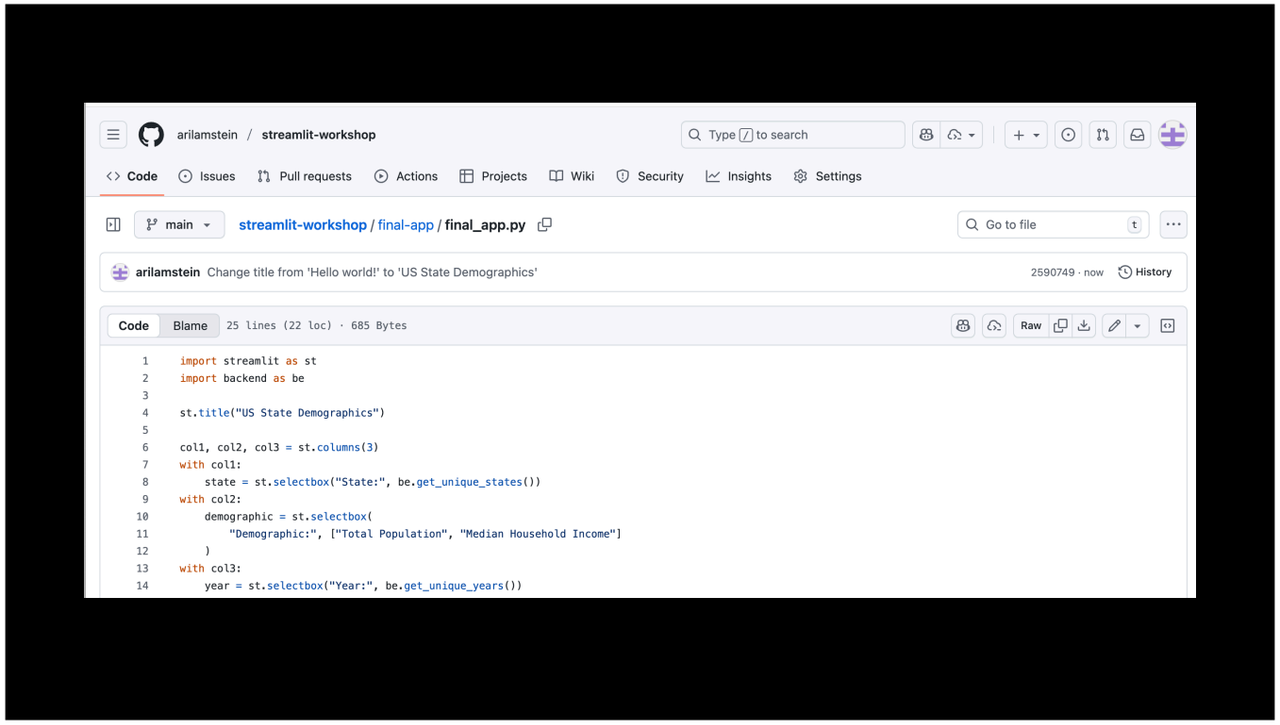
Deployment

- Get the code into Github
- Tell Streamlit Community Cloud about the Repo
- Notes about the deployed app

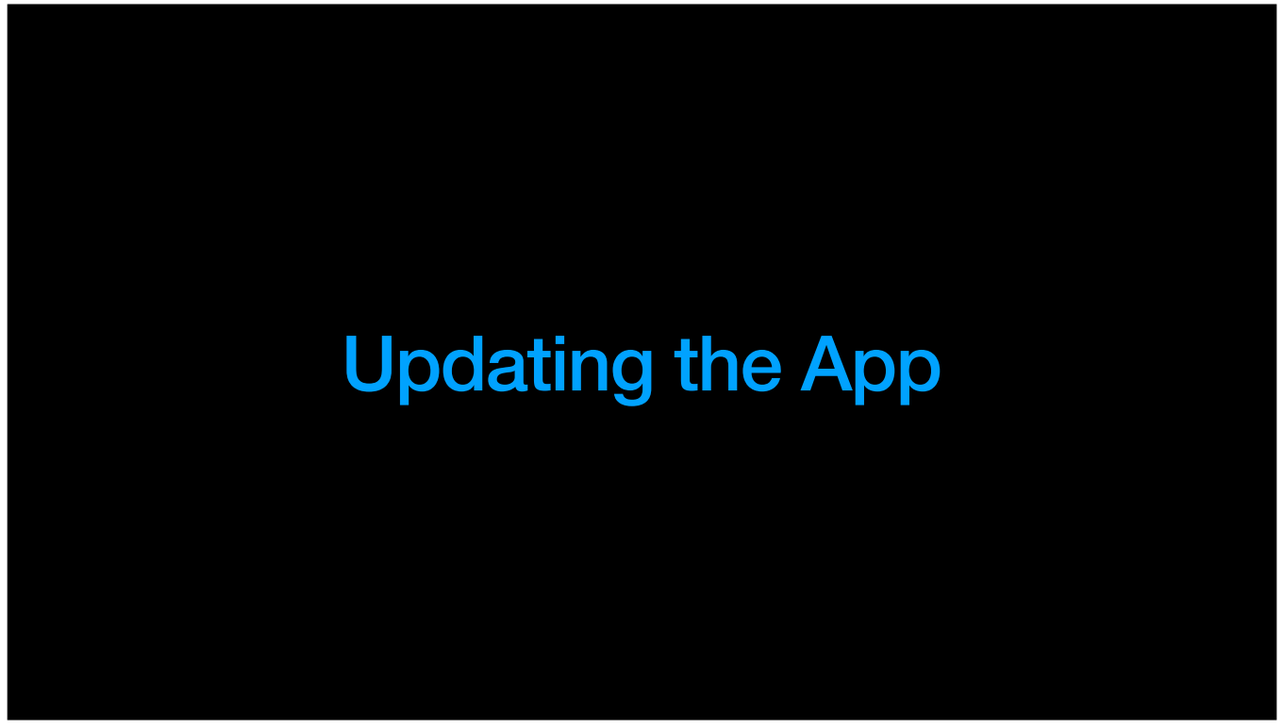
Before closing this section, let me share information that will help you manage your deployed app.



There are some other things you should learn about streamlit cloud.
In the upper right are a few buttons and icons. If you click the "GitHub" icon, it will take you to the GitHub repo for the app. Click it.



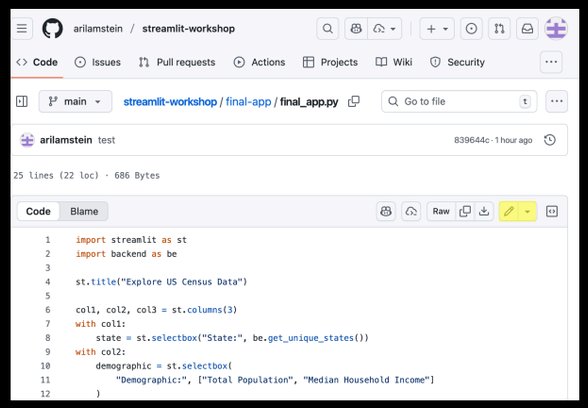
It should take you to the file that runs your app. For my app, that's final-app/final_app.py



The app will automatically update each time you push a change to the repo. You don't have to do anything - this will happen automatically. Let's walk through an example now.

Enter Edit Mode

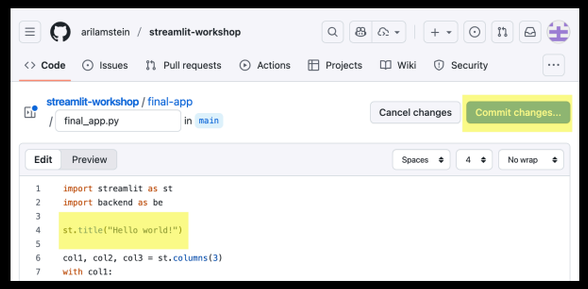
Click the "Pencil" icon next to the file



To start editing the file, click the pencil icon next on the right side of the screen. It's right above the file.

Edit the File

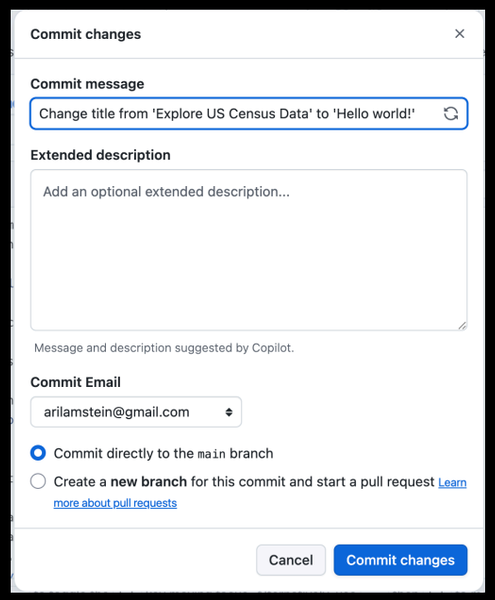
- Change the title to "Hello world!"
- Click "Commit changes"



After you click the pencil, you wind up in a screen to edit the file. Change the title to say something like "Hello world!". Then click "commit changes".

Commit your changes

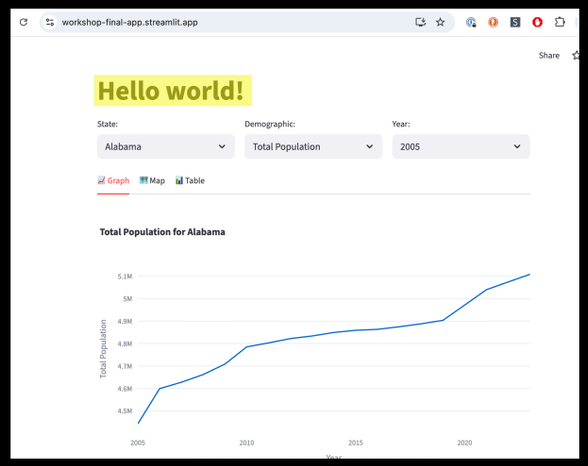
Click "Commit changes" on the commit screen.



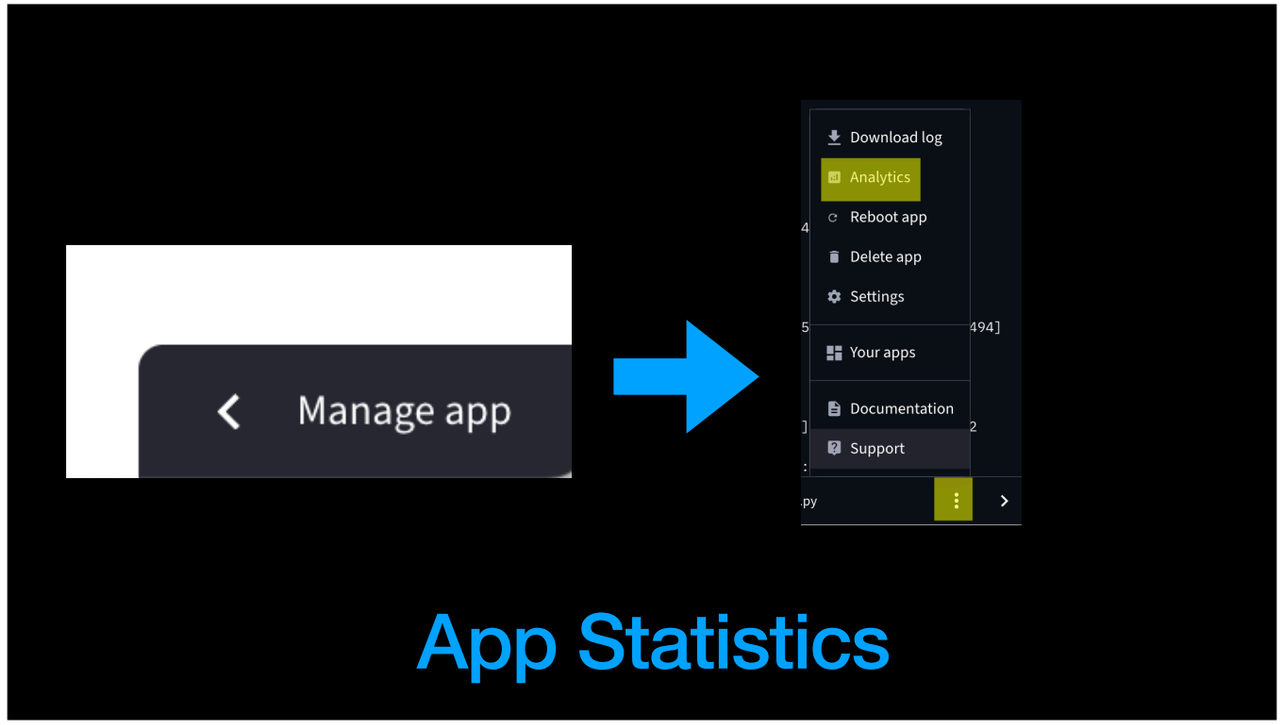
After you click "commit changes" a separate confirmation page will pop up. Click "commit changes" on the bottom.

View the app

Confirm that the changes took place in the app



Now go back to the app. You should see the title change.

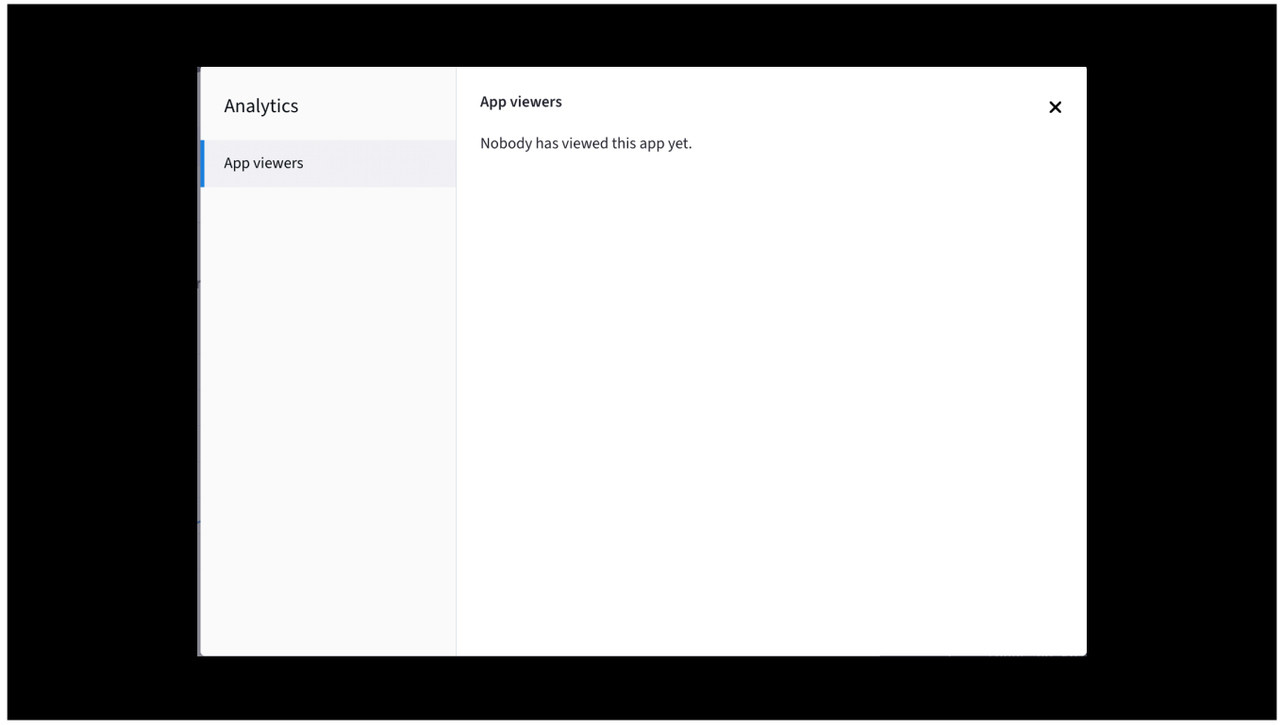


At some point you might want to know how many people have viewed your app.

In the lower right hand corner, click "Manage app".

Then click the icon with 3 vertical dots.

Then click "Analytics".



Because the app is new, no one has viewed it yet.

Deployment

- Get the code into Github
- Tell Streamlit Community Cloud about the Repo
- Notes about the deployed app

This wrapped up what I wanted to say about deploying your streamlit apps. First you need to get the code into github. Then you need to Streamlit's Community Cloud that you want to deploy the app.

Course Agenda
(Big Picture)

1. Session 1: Setup & Basics
2. Session 2: "The Loop": Input & Graphics
3. Session 3: Organization: UI & Code
4. Session 4: Deployment

Exercise: What is a Data App?

- Introduce yourself to your neighbor.
- Ask them 2 questions:
  - *What do you already know about Data Apps, Python and Streamlit?*
  - *What do you want to build after completing this course?*

Focus on the last question: now that you've completed this course, what do you want to build?